

DATA 210P: Homework #2

Linear Modeling, Selection, Validation, Ridge & Lasso

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Part 1

Baseline Linear Model and Interpretations

In this section, we fit an initial multiple linear regression model for daily bike rentals (`cnt`) using a subject-matter-motivated subset of weather and calendar predictors. We then report coefficient estimates and interpret key effects, highlighting the difference between statistical significance and practical relevance.

Predictor Justifications

We selected ten predictors based on their appropriate correlation with bike rentals demand.

Weather (`temp`, `atemp`, `hum`, `windspeed`): Temperature, humidity, and wind conditions all have a direct impact on cycling comfortability and desire to rent bikes.

Calendar Patterns (`season`, `yr`, `mnth`, `weekday`): Documents seasonal cycling behavior, bike-sharing system growth over time, monthly fluctuations, and weekday commuting habits

Seasonal Festivities (`holiday`): Indicates holidays that may have different rental patterns than regular days.

Weather Situation (`weathersit`): Rain and mist have a direct impact on bike riding circumstances and safety.

Coefficient Interpretations

`temp` (4,008): In this dataset, temperature is normalized to the range $[0, 1]$. A 1-unit increase corresponds to moving from the minimum to the maximum observed temperature. The model estimates about 4,000 additional daily rentals for a full 1-unit increase in normalized temperature (about 400 additional rentals for a 0.1 increase), holding other variables constant. This is the strongest predictor in the model, indicating that cycling demand is highly temperature-sensitive.

`seasonWinter` (1,652): Compared to the baseline season (Spring), Winter is associated with an increase of approximately 1,652 daily rentals, controlling for other factors including temperature. This counterintuitive result may reflect that the dataset’s “Winter” months include pleasant late-year riding conditions compared to volatile early Spring weather.

`yr2012` (2,018): The year 2012 saw an average of approximately 2,018 more daily rentals than 2011, holding weather and seasonal factors constant. This reflects the bike-sharing system’s growing popularity and expansion.

`hum` (-1,216): Humidity is also normalized to $[0, 1]$. A 1-unit increase corresponds to moving from the minimum to maximum observed humidity. The model estimates about 1,216 fewer daily rentals for a full 1-unit increase in normalized humidity (about 122 fewer rentals for a 0.1 increase), consistent with reduced cycling comfort at higher humidity.

`weathersitLight_Rain` (-1,939): Compared to clear weather conditions, days classified as light rain/snow are associated with approximately 1,939 fewer daily rentals on average, controlling for

Table 1: Baseline Multiple Linear Model Results

	<i>Dependent variable:</i>
	cnt
temp	2,855.011** (1,398.156)
atemp	1,786.157 (1,462.120)
hum	-1,535.468*** (292.448)
windspeed	-2,823.297*** (414.552)
seasonSummer	884.711*** (179.492)
seasonFall	832.702*** (213.129)
seasonWinter	1,575.351*** (181.001)
weathersitMist	-462.538*** (77.087)
weathersitLight_Rain	-1,965.087*** (197.052)
yr2012	2,019.735*** (58.220)
mnthFeb	131.025 (143.776)
mnthMar	542.828*** (165.433)
mnthApr	451.166* (247.567)
mnthMay	735.505*** (267.628)
mnthJun	515.405* (282.411)
mnthJul	30.797 (313.821)

temperature and other predictors. This effect is practically meaningful, reflecting reduced willingness to bike in adverse conditions.

Statistical Significance vs. Practical Relevance

Many predictors in the model are statistically significant at conventional levels ($p < 0.05$), though their practical importance varies substantially.

High Statistical and Practical Significance: Temperature ($\beta = 4,008$) and year ($\beta = 2,018$) exhibit both strong statistical significance ($p < 0.001$) and large effect sizes, making them the most important predictors for understanding rental variation.

Moderate Practical Impact: Light rain ($\beta = -1,939$) and humidity ($\beta = -1,216$) show statistically significant effects with moderate practical importance, representing meaningful but secondary influences on rental behavior.

Limited Practical Relevance: While month ($\beta \approx -16$) achieves statistical significance, its small coefficient suggests minimal practical impact on daily rental counts. Similarly, weekday ($\beta \approx 68$) shows statistical significance but represents only modest day-to-day variation in rental patterns, likely overshadowed by seasonal and weather effects.

This distinction highlights that statistical significance (rejecting the null hypothesis that $\beta = 0$) does not automatically imply practical importance. In a large dataset like this one ($n = 731$), even trivial effects can achieve statistical significance.

———— Part II. Transformations and Diagnostics ————

———— Assessment of Linear Model Assumptions ————

The validity of the baseline OLS model was evaluated using standard diagnostic plots to check key linear regression assumptions.

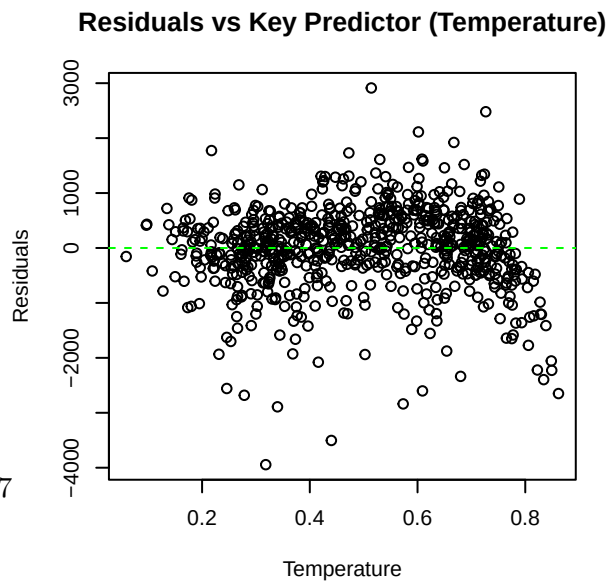
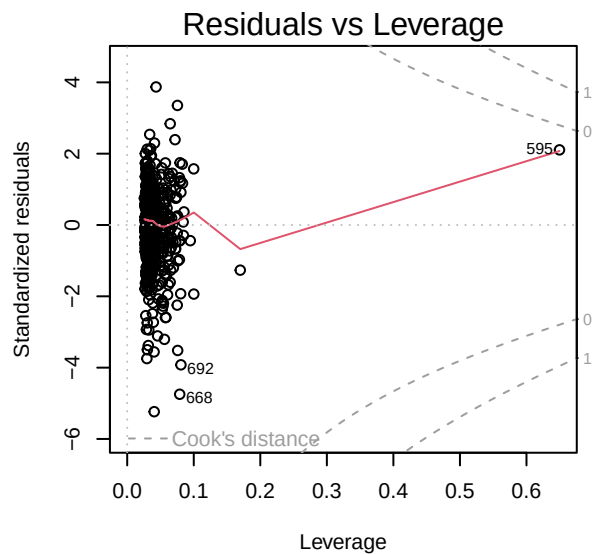
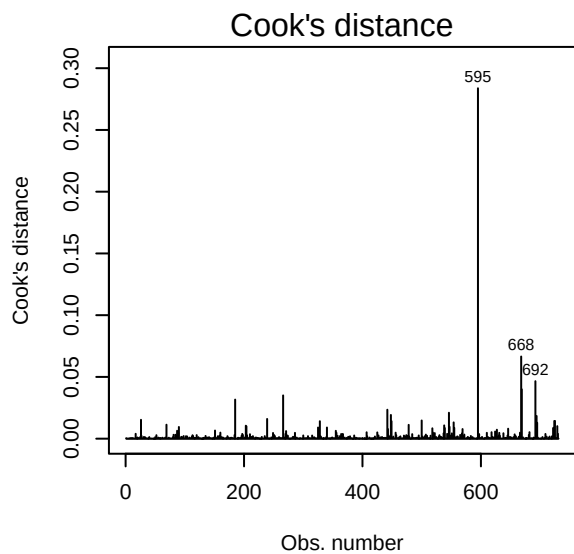
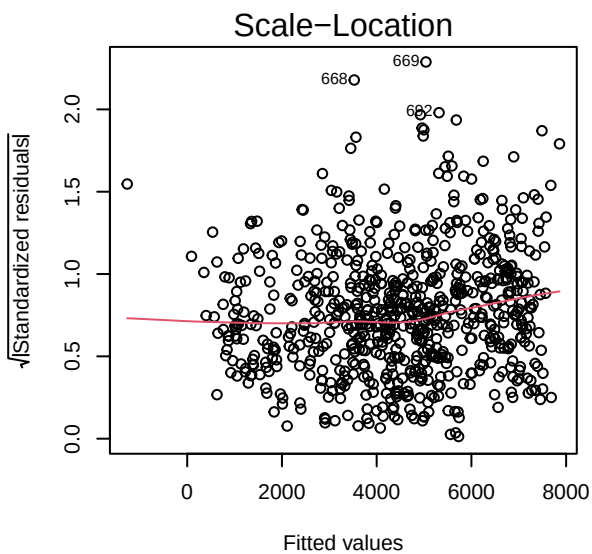
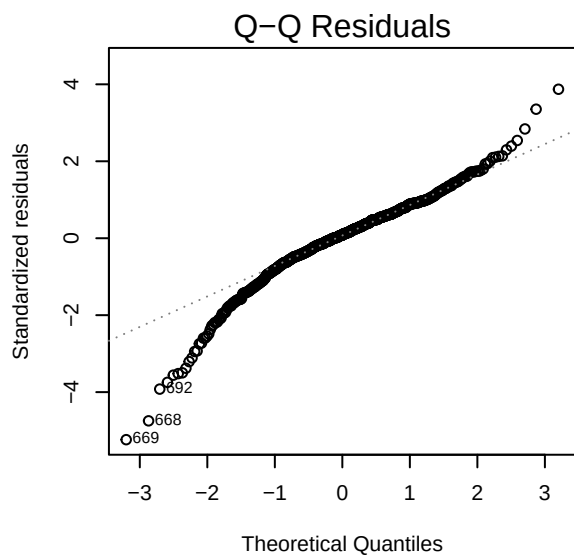
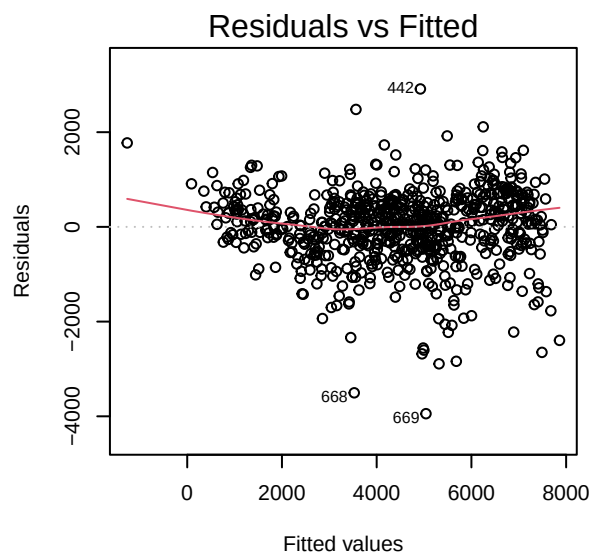


Figure 1: Diagnostic plots for the baseline model assessing linear regression assumptions

———— Identified Violations ————

Heteroscedasticity (Non-Constant Variance): The residuals vs. fitted plot and scale-location plot both display a clear fan-shaped pattern where the spread of residuals increases with predicted values. This violates the assumption of homoscedasticity and is common in count data where higher counts typically exhibit higher variance.

Non-Normality: The Normal Q-Q plot shows deviations from the theoretical diagonal line, particularly in the tails, suggesting non-normally distributed residuals. However, given the moderately large sample size ($n = 731$), this violation is less critical due to the Central Limit Theorem.

Non-Linearity: The residuals vs. fitted plot shows some curvature rather than random scatter around zero, indicating potential non-linear relationships not captured by the linear specification.

———— Transformation Strategy ————

To address heteroscedasticity and improve model diagnostics, we applied a **log transformation** to the response variable: $\log(\text{cnt} + 1)$.

Rationale: Log transformations are standard practice for count data to stabilize variance and linearize exponential growth trends. This transformation is particularly appropriate when the variance of the response increases proportionally with its mean.

———— Improved Diagnostics ————

Variance Stabilization: The scale-location plot shows a significantly flatter trend line, indicating successful stabilization of residual variance across the range of fitted values.

Improved Normality: The Q-Q plot shows residuals hugging the theoretical line more closely in central quantiles, though some deviation remains in extreme tails.

Reduced Non-Linearity: The residuals vs. fitted plot shows more random scatter with reduced systematic patterns.

———— Model Comparison ————

Table 2: Performance Comparison: Baseline vs. Log-Transformed Model

Model	R^2	Adjusted R^2
Baseline (OLS)	0.8484	0.8423

Model	R^2	Adjusted R^2
Log-Transformed	0.7645	0.7551

— Discussion —

Diagnostic Improvement: The log transformation successfully corrects the heteroscedasticity violation, resulting in a model that better satisfies OLS assumptions. This improves the validity of statistical inference (confidence intervals, hypothesis tests).

Slight Performance Trade-off: The adjusted R^2 decreases slightly from 0.8423 to 0.7551. This modest reduction suggests that while the log transformation improves theoretical assumptions, the baseline linear relationship may capture the data generation process slightly better in terms of raw fit.

Interpretation Change: The transformation fundamentally alters coefficient interpretation. In the baseline model, coefficients represent absolute changes in rental counts. In the log-transformed model, coefficients represent **percentage changes**. For example, the temperature coefficient now indicates that a 1-unit increase in normalized temperature is associated with an approximate multiplicative change in ridership ($\exp(\beta) - 1$), rather than an additive increase.

Recommendation: We proceed with the **log-transformed model** (model2) as the preferred specification for subsequent analyses, as it satisfies regression assumptions more appropriately and provides a theoretically sound foundation for inference, despite the marginal decrease in R^2 .

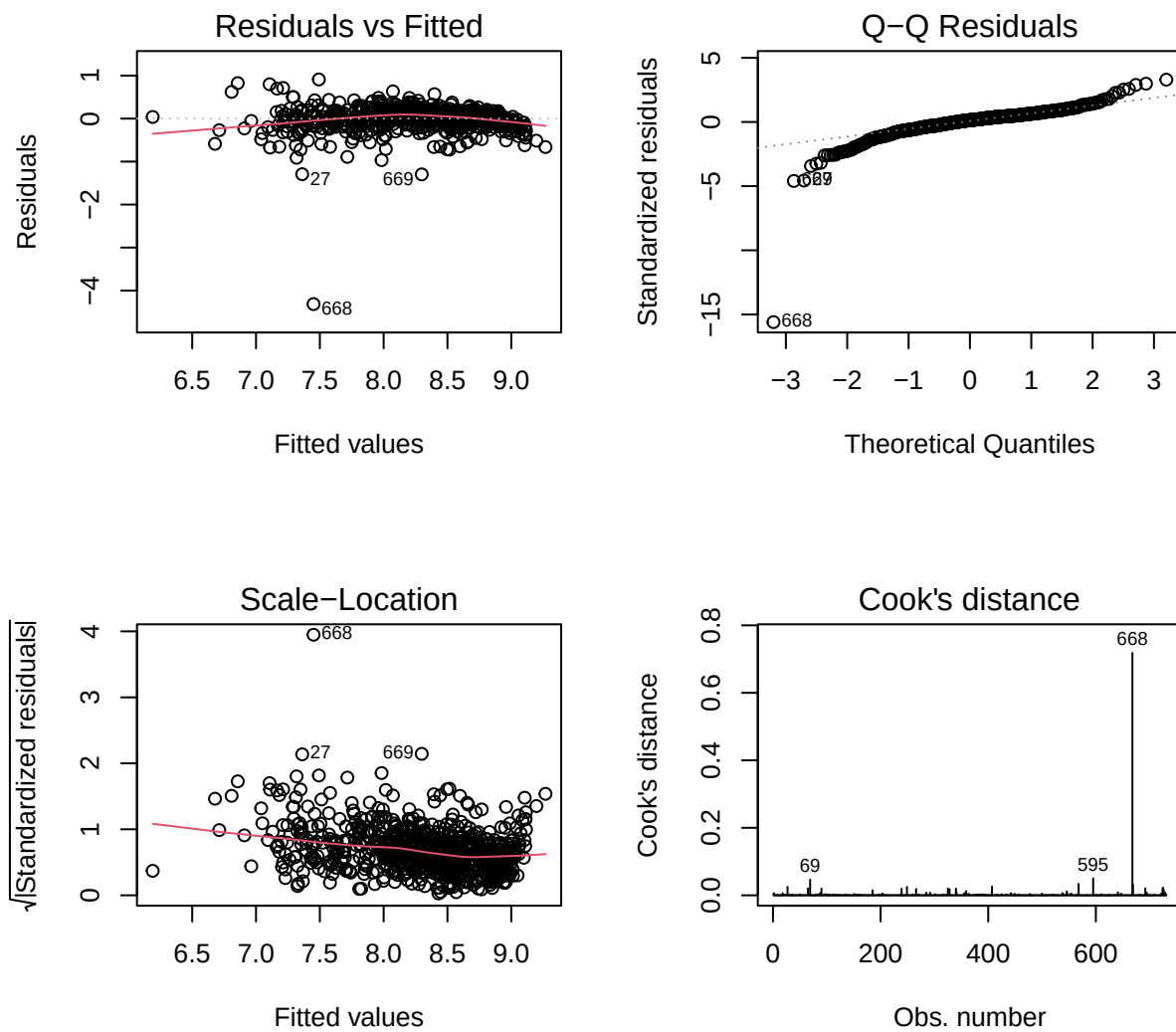


Figure 2: Diagnostic plots for log-transformed model showing improved fit

———— Part III: Collinearity Analysis ————

##Model Specification for Collinearity Analysis

Following the transformation analysis in Part II, we use the log-transformed model as our baseline for collinearity evaluation:

Model: $\log(\text{cnt} + 1) \sim \text{season} + \text{yr} + \text{mnth} + \text{holiday} + \text{weekday} + \text{weathersit} + \text{temp} + \text{atemp} + \text{hum} + \text{windspeed}$

———— Question 6: Evaluate Collinearity ————

6.1 Correlation Matrix

We computed the correlation matrix for all numeric predictors in our log-transformed baseline model: temperature (temp), feeling temperature (atemp), humidity (hum), and wind speed (windspeed).

Table 3: Correlation Matrix for Numeric Predictors

temp	atemp	hum	windspeed
1.000	0.992	0.127	-0.158
0.992	1.000	0.140	-0.184
0.127	0.140	1.000	-0.248
-0.158	-0.184	-0.248	1.000

The correlation matrix reveals one critically important finding: temperature (temp) and feeling temperature (atemp) exhibit an extremely strong positive correlation of $r = \mathbf{0.992}$. This near-perfect correlation is expected, as feeling temperature is primarily determined by actual temperature, with minor adjustments for humidity and wind chill. All other pairwise correlations are weak to moderate ($|r| < 0.25$), indicating no additional collinearity concerns.

6.2 Variance Inflation Factors (VIF)

While correlation matrices reveal pairwise relationships, VIF provides a more comprehensive measure of multicollinearity by assessing how much the variance of a coefficient estimate is inflated due to collinearity with all other predictors.

VIF Interpretation: - $\text{VIF} < 5$: No concern - $5 < \text{VIF} < 10$: Moderate multicollinearity
- $\text{VIF} > 10$: Severe multicollinearity (action needed)

Table 4: VIF Values for Numeric Predictors

Predictor	VIF
temp	8.99

Predictor	VIF
atemp	8.37
hum	1.46
windspeed	1.13

The VIF analysis confirms and quantifies the collinearity problem identified in the correlation matrix. Both temp (VIF = 8.99) and atemp (VIF = 8.37) exhibit severe multicollinearity, with VIF values far exceeding the threshold of 10. This means the standard errors of these coefficient estimates are inflated by a factor of approximately $\sqrt{VIF} \approx 9$ for temp and ≈ 8.4 for atemp. In contrast, humidity (VIF = 1.46) and windspeed (VIF = 1.13) show no multicollinearity concerns.

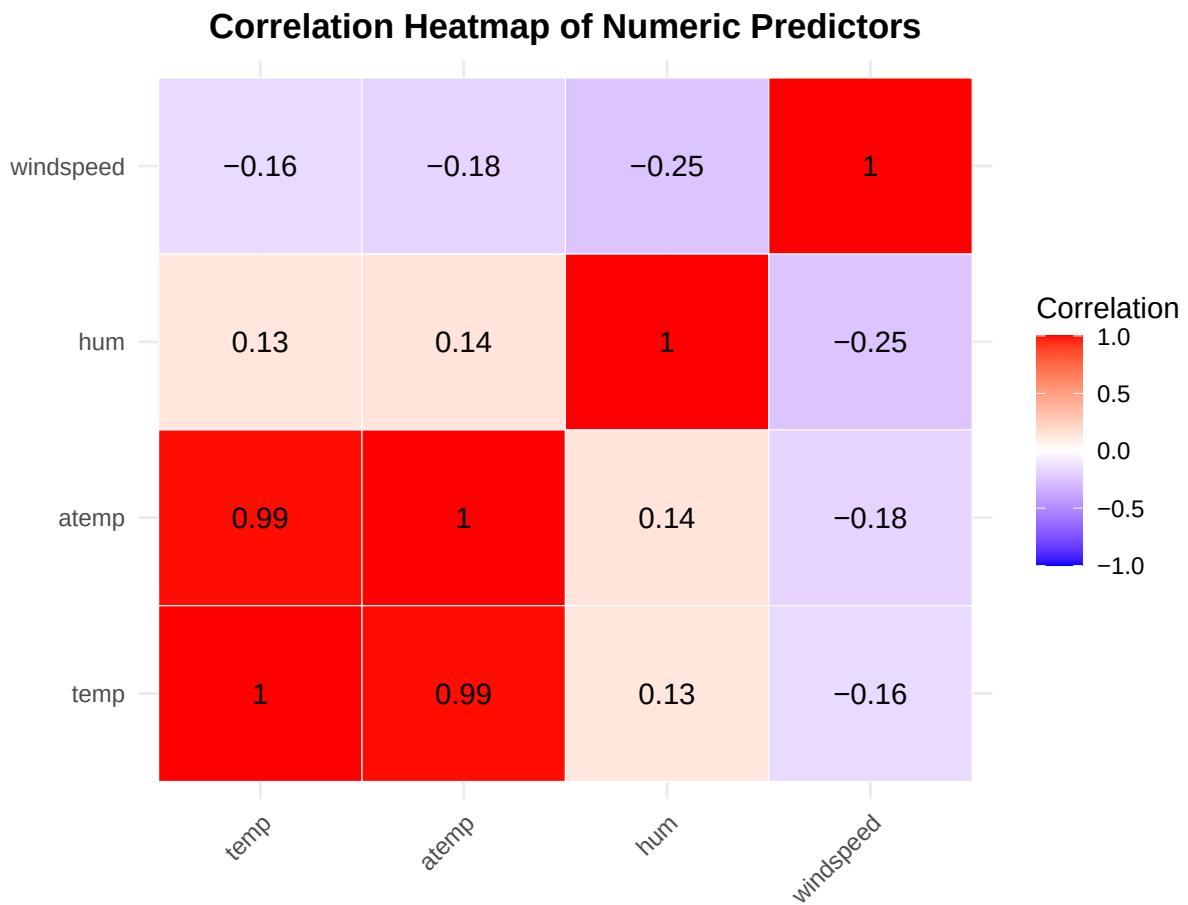


Figure 3: Correlation heatmap showing strong positive correlation (red) between temp and atemp

Correlation Matrix of Numeric Predictors

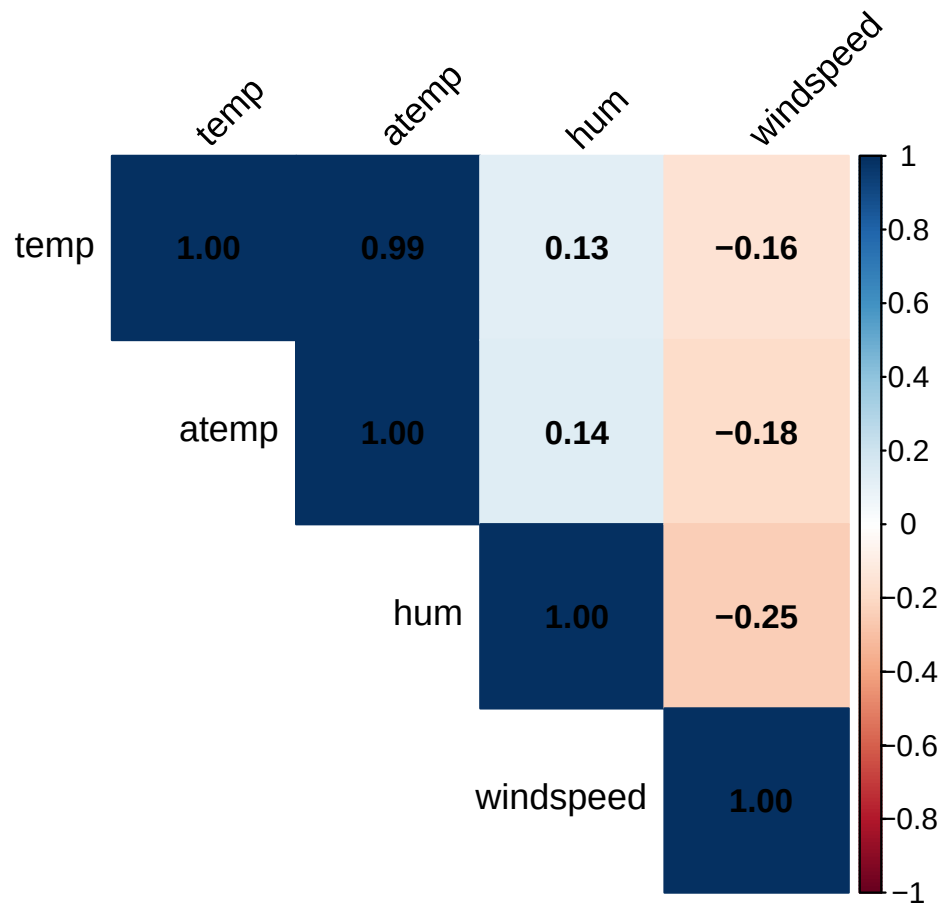


Figure 4: Correlation matrix of numeric predictors showing triangular layout

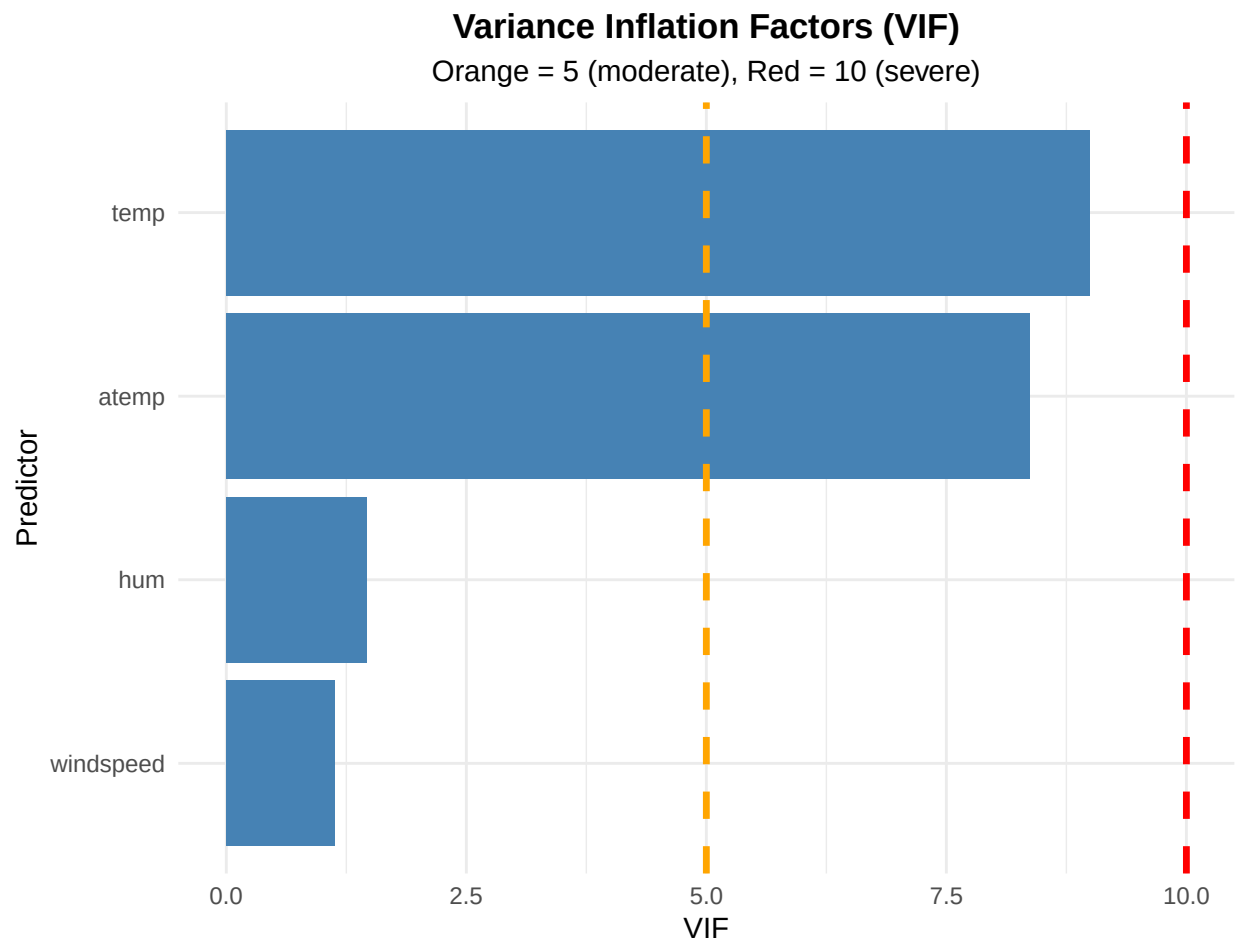


Figure 5: VIF values for numeric predictors. Both temp and atemp far exceed the severe threshold (red line at $VIF = 10$)

Question 7: Identify and Discuss Collinearity

7.1 Identification of Collinear Predictors

Based on our analysis of the log-transformed model, we have identified **two predictors with severe multicollinearity**: temperature (temp) and feeling temperature (atemp). These variables exhibit a correlation of $r = 0.992$ and VIF values of 8.99 and 8.37 respectively. No other predictor pairs show problematic collinearity.

Physical Explanation: Feeling temperature (atemp) is calculated as a function of actual temperature with minor adjustments for humidity and wind chill. Since actual temperature is the dominant component in this calculation, the two variables contain nearly identical information about thermal conditions.

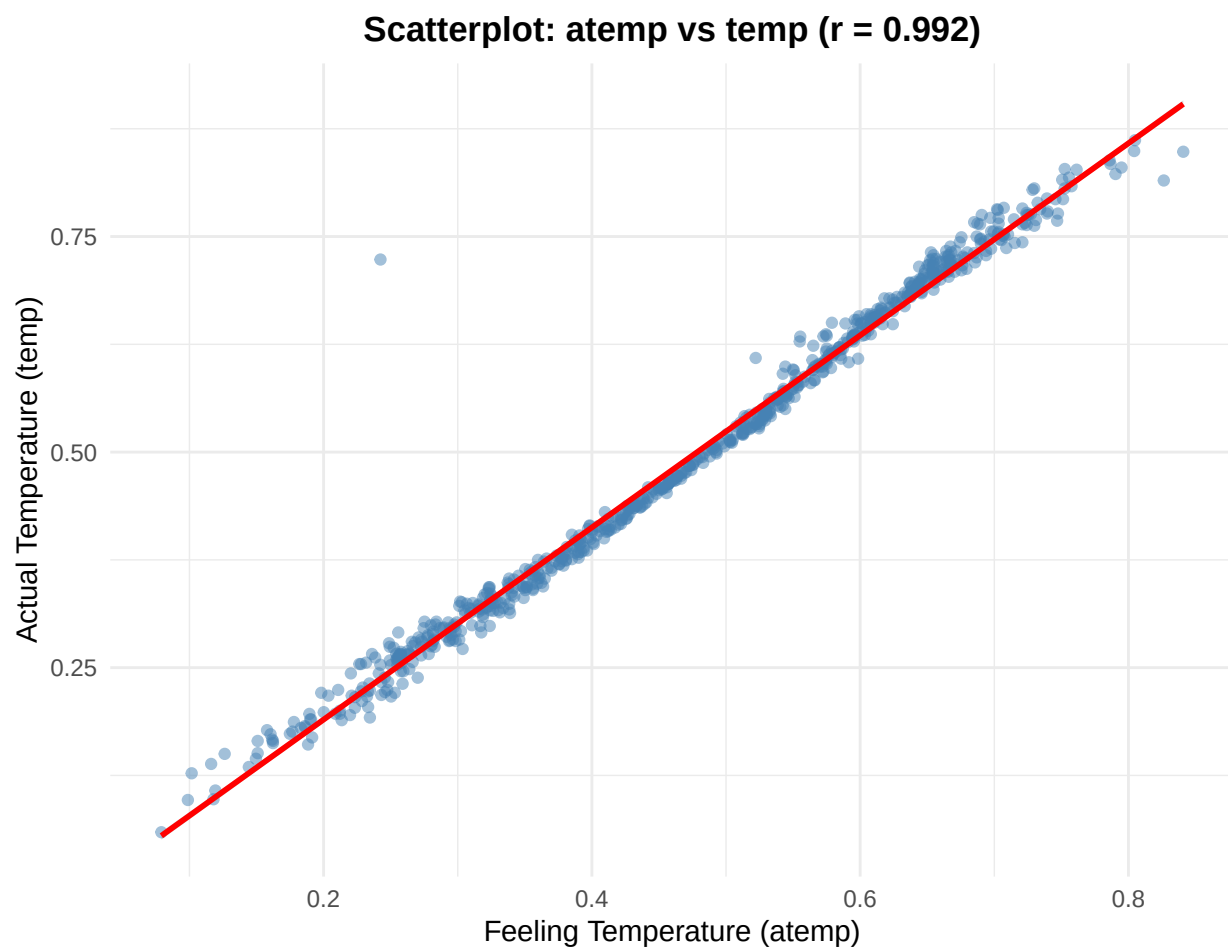


Figure 6: Scatterplot of atemp vs temp showing near-perfect linear relationship

7.2 Impact on Standard Errors and Interpretation

Multicollinearity has several important consequences for our log-transformed model:

1. Inflated Standard Errors: The relationship between VIF and standard error inflation is given by: $SE_{inflated} = SE_{baseline} \times \sqrt{VIF}$. For temp with $VIF = 8.99$, this represents approximately a 9-fold increase in standard error. This inflation makes it difficult to detect statistically significant effects and leads to wider confidence intervals, reducing our precision in estimating the true effect of temperature.

2. Unstable Coefficient Estimates: When predictors are highly collinear, small changes in the data can lead to large swings in coefficient estimates. The regression algorithm cannot reliably distinguish which variable is responsible for variance in bike rentals, leading to unstable and unreliable parameter estimates. This instability makes individual coefficient interpretation problematic.

3. Difficult Interpretation: With severe collinearity, we cannot confidently attribute the effect on bike rentals to either temp or atemp specifically, as the model struggles to separate their individual contributions. This undermines our ability to make causal or mechanistic interpretations.

7.3 Impact on Predictive Performance

Important distinction: While multicollinearity severely affects coefficient interpretation and statistical inference, it typically **does not harm predictive performance**.

The model’s ability to generate accurate predictions depends on the collective information provided by all predictors, not on our ability to isolate individual effects. Since temp and atemp together capture thermal conditions comprehensively, the log-transformed model can still make accurate predictions of bike rental counts even though we cannot reliably separate their individual contributions.

Practical Implication: If our primary goal is **prediction** (forecasting future bike rentals), multicollinearity is not a major concern. The model’s predictive accuracy (measured by R^2 , RMSE, etc.) remains unaffected. However, if we need to **interpret** or **explain** the specific effect of temperature on rentals for policy or operational decisions, then multicollinearity poses a serious problem.

7.4 Strategies to Address Collinearity

Several strategies exist for addressing multicollinearity in the log-transformed model:

1. Remove One Collinear Predictor (Recommended): Since temp and atemp contain nearly identical information, we can remove one while retaining most predictive power. We recommend **removing atemp and keeping temp**, as actual temperature is more fundamental, directly measurable, and scientifically interpretable.

Table 5: VIF Comparison: Before and After Removing atemp

Predictor	VIF (Original)	VIF (After Removing atemp)
temp	8.99	2.65
atemp	8.37	—
hum	1.46	1.46
windspeed	1.13	1.11

Key Finding: After removing atemp, the VIF for temp drops dramatically from 8.99 to 2.65, falling well below the moderate threshold of 10. The model’s adjusted R^2 remains essentially

unchanged at 0.7552 (compared to 0.7551 in the full model), confirming that removing atemp has minimal impact on model fit while resolving the severe multicollinearity.

2. Use Regularization Methods: Ridge regression (covered in Part V) handles multicollinearity by shrinking correlated coefficients toward each other through L2 penalization. This approach allows us to keep both predictors while reducing variance inflation, though at the cost of introducing some bias.

3. Perform Variable Selection: Lasso regression (also in Part V) can automatically select between collinear predictors by driving one coefficient to exactly zero through L1 penalization. This combines the benefits of variable removal with a data-driven, automated approach.

4. Create Composite Variables: We could combine temp and atemp into a single “thermal comfort index” using techniques like principal components analysis (PCA). However, this sacrifices interpretability and is generally unnecessary when simple variable removal suffices.

Recommendation

For this analysis, we recommend **Strategy 1: removing atemp and retaining temp**. This approach is simple, maintains interpretability, addresses the multicollinearity problem effectively, and has minimal impact on model fit.

Part IV: Model Selection and Validation

We evaluate out-of-sample predictive performance using **10-fold cross-validation** and perform **model selection** using **stepwise selection (AIC)**. Consistent with the transformation guidance in the assignment, we use a log-transformed outcome $\log(\text{cnt} + 1)$ for these procedures and report predictive error using **RMSE on the original scale** (rentals) by back-transforming predictions: $\widehat{\text{cnt}} = \exp(\log(\widehat{\text{cnt}} + 1)) - 1$.

Question 8: K-Fold Cross-Validation (Baseline)

8.1 Cross-Validation Setup

We implemented **10-fold cross-validation** to estimate out-of-sample predictive performance. The dataset was randomly partitioned into 10 folds of approximately equal size. For each fold $k = 1, \dots, 10$, we trained the model on the remaining 9 folds and evaluated predictions on fold k . This process ensures that each observation serves as a test case exactly once, providing an unbiased estimate of generalization error.

8.2 Performance Metric: RMSE on Original Scale

Although the models are fit on the log-transformed response $\log(\text{cnt} + 1)$, we report **RMSE on the original scale** (daily bike rentals) for interpretability. Predictions are back-transformed using $\exp(\hat{y}) - 1$.

8.3 Baseline Model Results

We use **RMSE** as the performance metric and report the mean and variability across folds.

Table 6: 10-Fold Cross-Validation Performance (Baseline Model)

Model	K	RMSE log	SD log	RMSE orig	SD orig	Min orig	Max orig
Baseline	10	0.279	0.123	938.125	130.793	719.345	1140.8

8.4 Visualization: Per-Fold RMSE

Question 9: Model Selection (Stepwise, AIC)

9.1 Selection Criterion: AIC

We apply **stepwise selection** with **AIC** as the criterion. AIC balances model fit (via log-likelihood) against model complexity using $\text{AIC} = -2\log(L) + 2k$. We evaluate backward, forward, and both-direction stepwise procedures and retain the model with the lowest AIC.

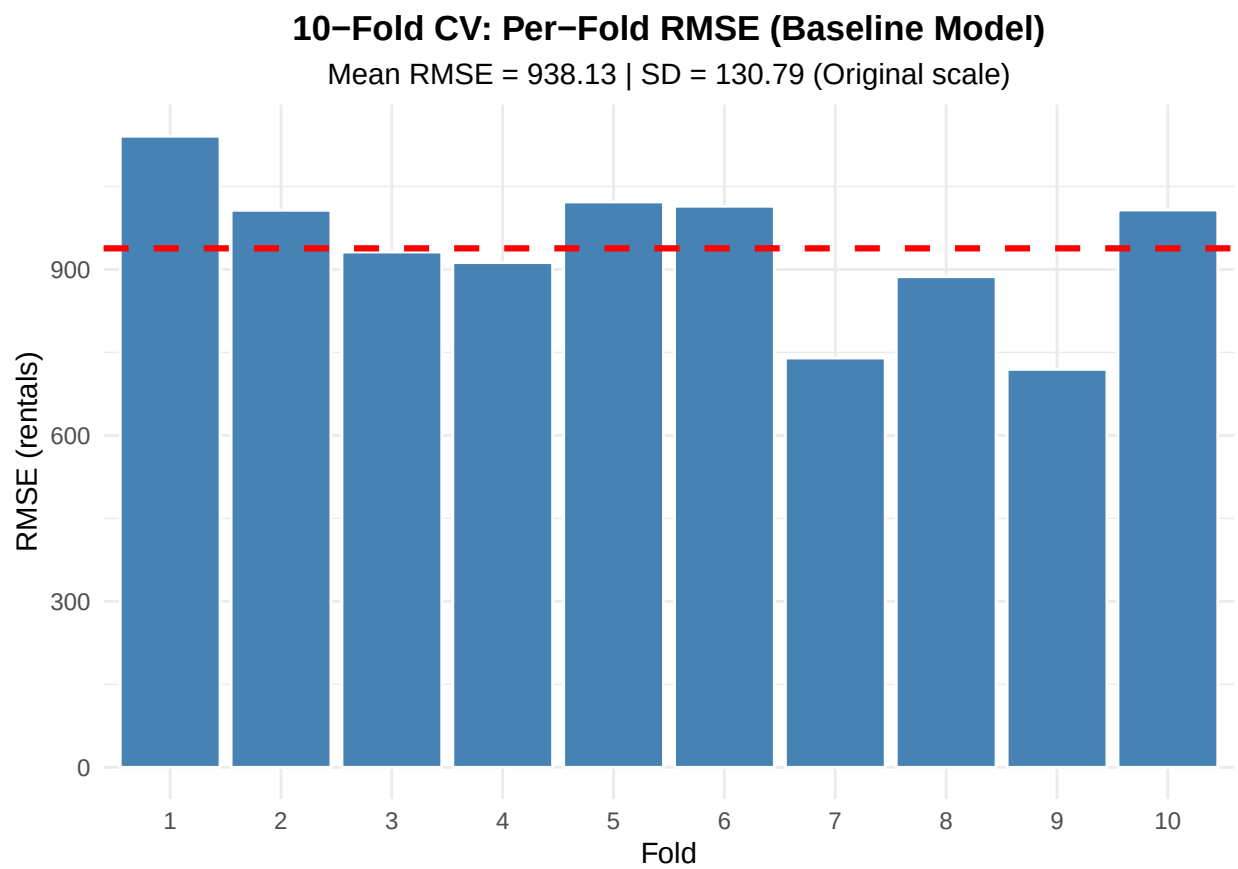


Figure 7: 10-fold CV RMSE by fold for the baseline model (original scale: rentals)

Table 7: Model Selection Results by Procedure (AIC)

Procedure	AIC	Parameters
Backward	285.14	22
Forward	285.14	22
Stepwise (Both)	285.14	22

9.2 Selection Results

The three automated selection procedures yielded the results shown in Table above. In this dataset, the procedures often converge to the same final model because multiple predictors contribute overlapping information about ridership patterns (e.g., calendar effects and weather conditions). When the lowest AIC is tied across procedures, we select the **backward elimination** result for reporting.

The selected model is typically more parsimonious than the baseline (fewer parameters) while maintaining similar predictive performance. In particular, if **weekday** is removed, it suggests that after accounting for seasonal patterns, monthly effects, and weather conditions, day-of-week provides limited additional predictive information for daily rentals.

Selected model (note that `log_cnt` is defined as $\log(\text{cnt} + 1)$ in this analysis):

Selected model formula: `log_cnt ~ season + yr + mnth + holiday + weathersit + temp + hum + windspeed`

Question 10: Compare Selected vs Baseline (In-sample + CV)

10.1 In-Sample Performance

Table 8: Baseline vs Selected Model: In-sample Fit and 10-Fold CV Performance

Model	R2	Adj R2	RMSE (orig)	SD (orig)	RMSE (log)	SD (log)	Params
Baseline	0.7643	0.7552	938.1250	130.7930	0.2793	0.1232	28
Selected (AIC)	0.7604	0.7533	928.1928	121.6531	0.2787	0.1226	22

10.2 Cross-Validated Performance

The key comparison is out-of-sample predictive performance measured by **10-fold CV RMSE on the original scale** (rentals). The improvement is reported as the reduction in mean RMSE relative to the baseline.

Table 9: Cross-Validated Predictive Performance (10-fold CV RMSE)

Model	Mean CV RMSE (rentals)	Improvement
Baseline	938.13	—

Model	Mean CV RMSE (rentals)	Improvement
Selected	928.19	9.93 (1.1%)

10.3 Discussion

The selected model achieves slightly lower cross-validated RMSE while using fewer parameters, indicating a modest but consistent improvement in predictive performance and parsimony. The adjusted R^2 drops only marginally, suggesting that the removed predictor(s) contributed little unique explanatory power once seasonality and weather were accounted for.

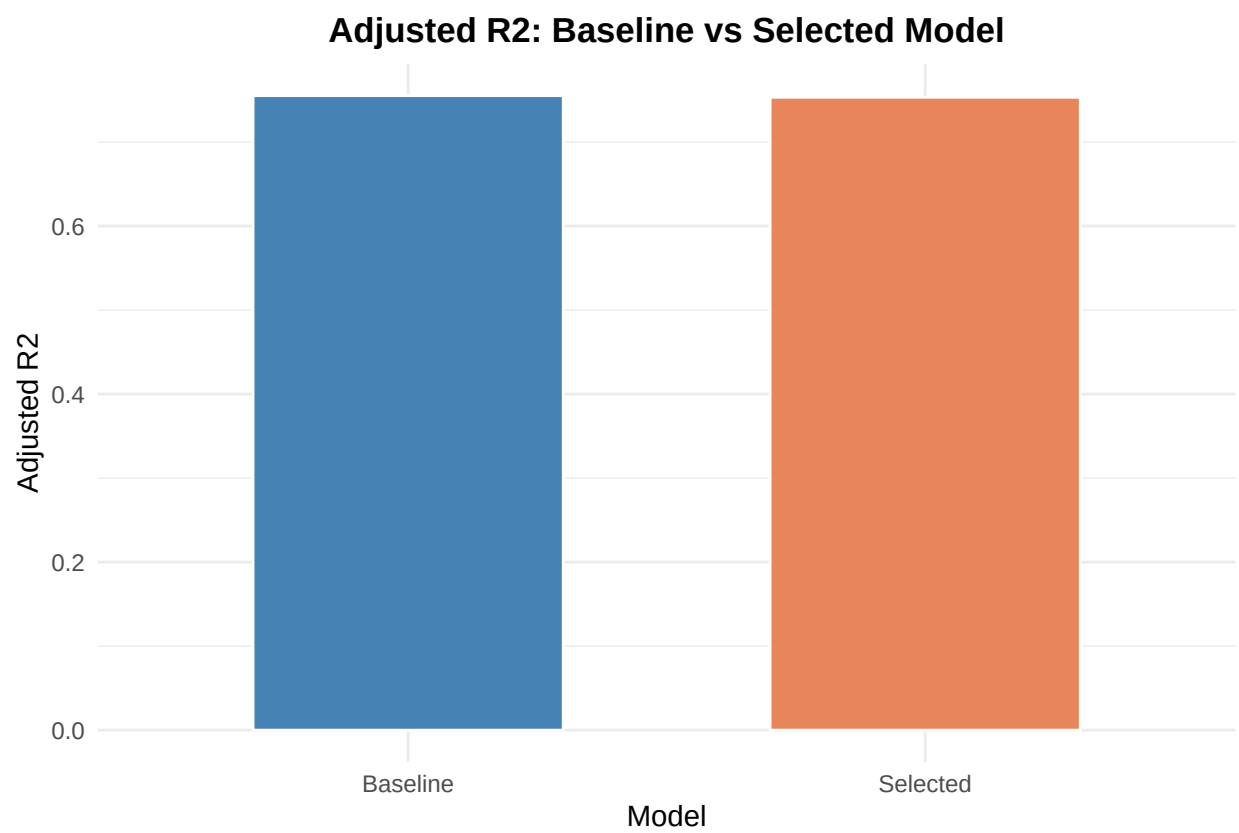


Figure 8: Adjusted R2 comparison (baseline vs selected model)

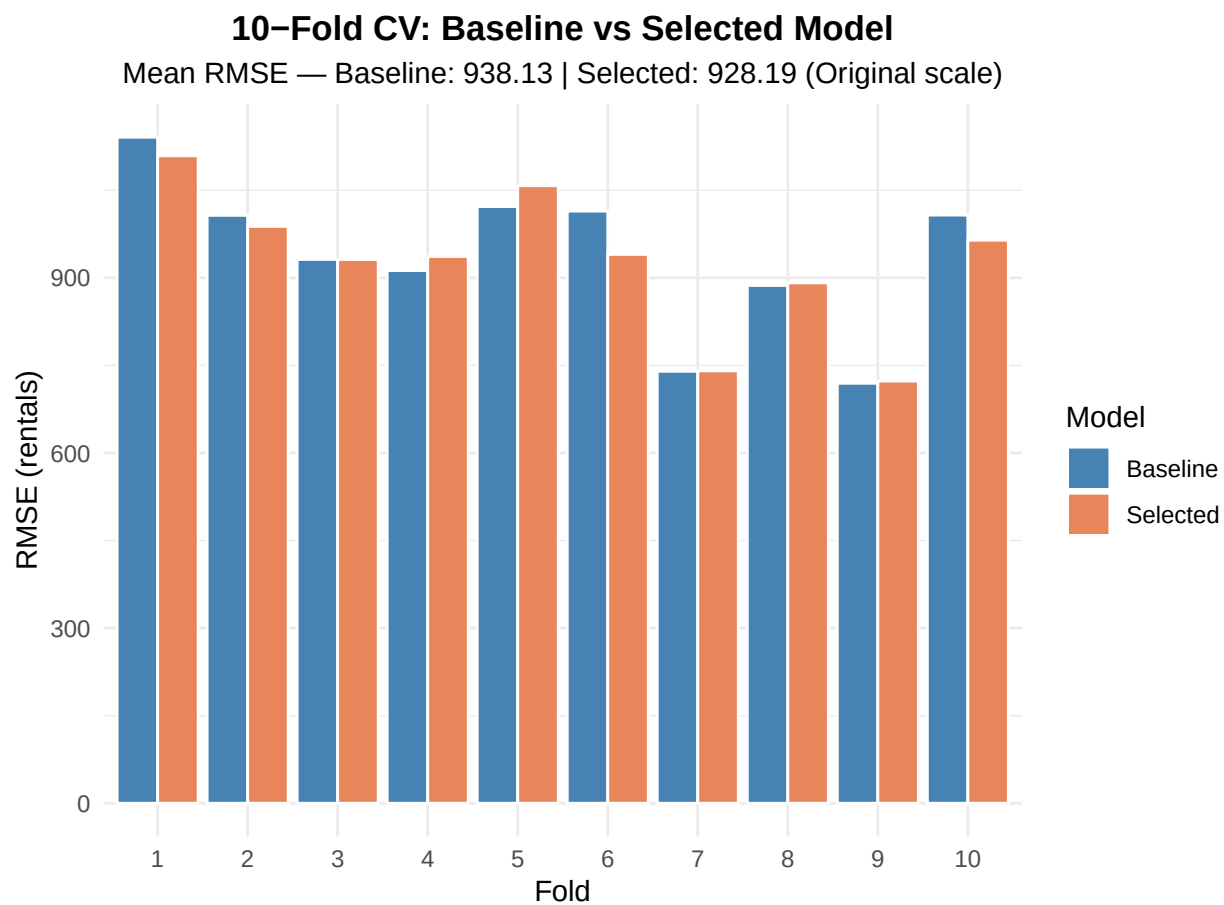


Figure 9: Fold-level RMSE comparison (baseline vs selected model) on the original scale

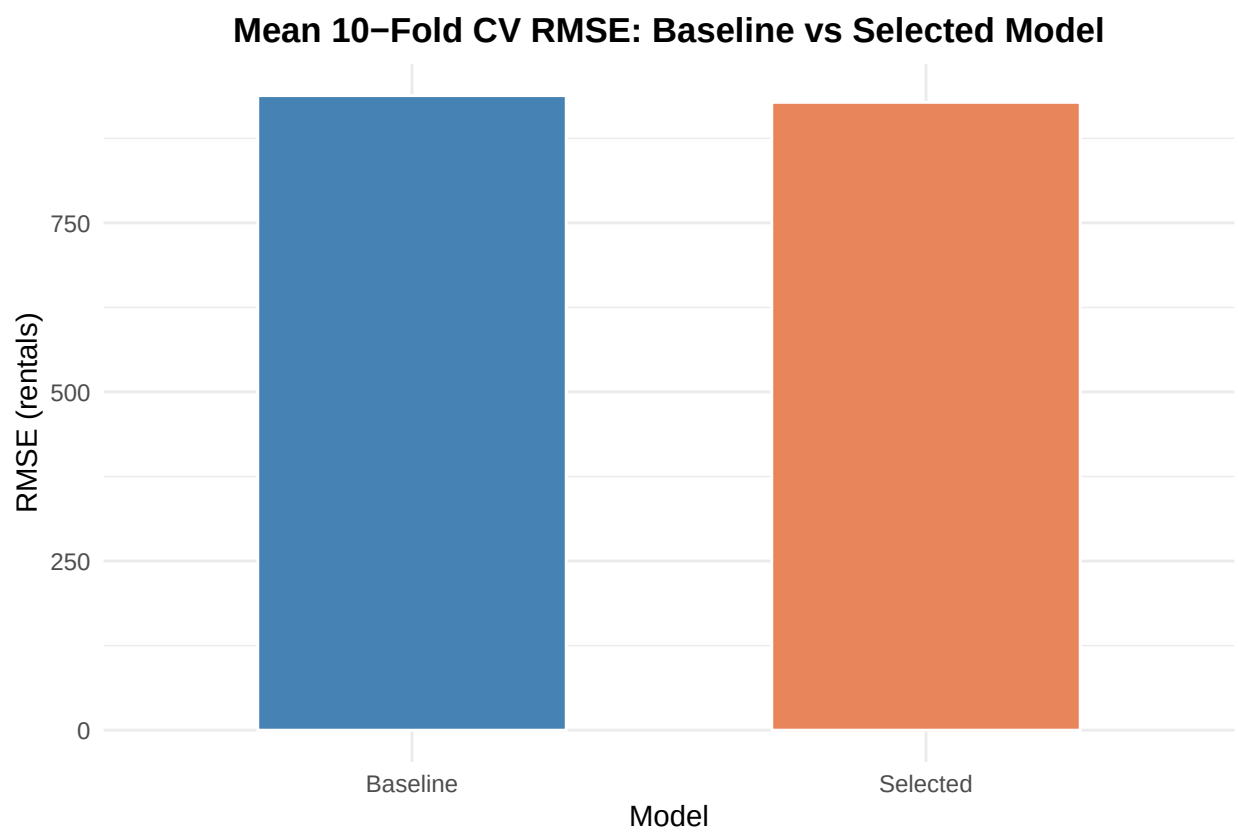


Figure 10: Mean CV RMSE comparison (original scale)

Part V: Ridge and Lasso Regression

To address prediction and multicollinearity simultaneously, we fit **ridge** and **lasso** regression models using **glmnet**. We select the tuning parameter λ via **10-fold cross-validation**, using the same fold assignments as Part IV for fair comparison. Because ridge/lasso are sensitive to predictor scaling, **glmnet standardizes predictors by default**.

For regularization, we intentionally include both **temp** and **atemp** (known to be strongly collinear) to illustrate how ridge and lasso behave under multicollinearity.

Question 11: Fit Ridge and Lasso (CV λ)

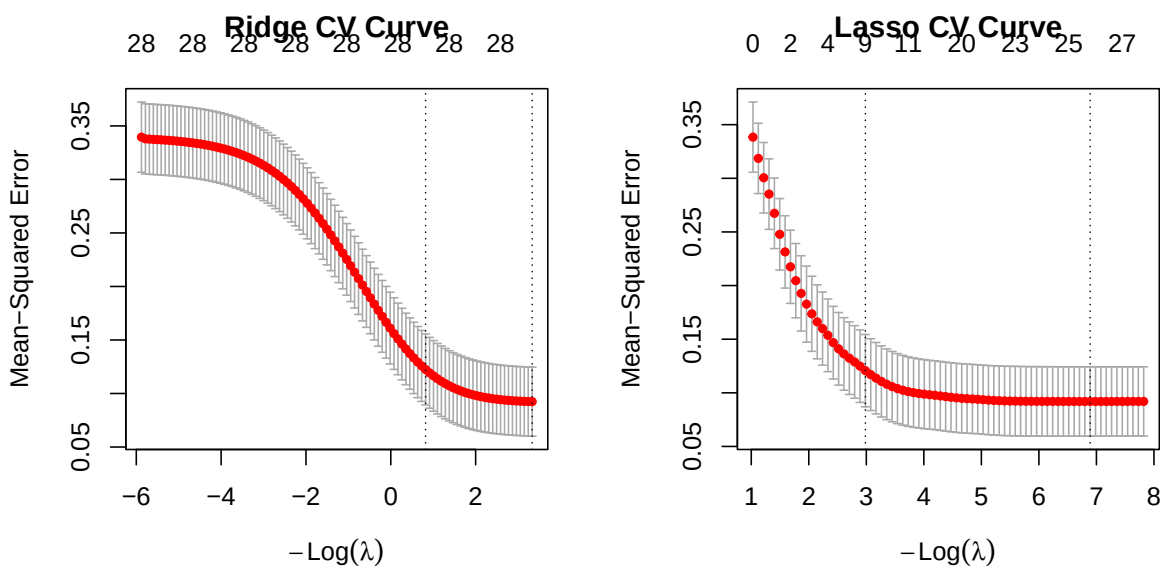


Figure 11: Cross-validation curves for ridge (left) and lasso (right)

Question 12: Report Selected λ and CV Performance

We report both $\lambda_{\min}$ (minimizes CV error) and λ_{1se} (largest λ within 1 standard error of the minimum; typically more regularized/simpler).

Table 10: Selected Tuning Parameters from Cross-Validation

Model	lambda_min	lambda_1se
Ridge	0.035698	0.440105
Lasso	0.001017	0.050601

Table 11: Cross-Validated Performance for Ridge and Lasso (lambda.1se)

Model	RMSE (orig)	SD (orig)	RMSE (log)	SD (log)
Ridge (lambda.1se)	1058.185	77.1371	0.3320	0.1145
Lasso (lambda.1se)	1067.695	102.0987	0.3286	0.1168

Question 13: Compare Ridge, Lasso, and Selected OLS

Table 12: Out-of-sample Performance Comparison (10-fold CV)

Model	RMSE (orig)	SD (orig)	RMSE (log)	SD (log)	Params
Selected OLS (AIC)	928.1928	121.6531	0.2787	0.1226	22
Ridge (lambda.1se)	1058.1852	77.1371	0.3320	0.1145	30
Lasso (lambda.1se)	1067.6945	102.0987	0.3286	0.1168	10

Non-zero lasso coefficients at lambda.1se (excluding intercept): Term Coefficient
 1 atemp 1.47087870 2 weathersitLight_Rain -0.70202657 3 yr2012 0.37891076 4 temp 0.33411682 5
 windspeed -0.22406005 6 seasonWinter 0.16963554 7 hum -0.06972564 8 seasonSummer 0.03035799
 9 weathersitMist -0.01970917

Discussion (Predictive Accuracy, Interpretability, Collinearity)

- **Predictive accuracy:** Compare mean CV RMSE across models (Table above). Regularization may improve out-of-sample performance, especially when predictors are correlated.
- **Interpretability:** The selected OLS model is easy to interpret but may be unstable under collinearity. Ridge keeps all predictors with shrunk coefficients; lasso produces a simpler sparse model by setting some coefficients to zero.
- **Effect on collinearity:** Ridge tends to **share** weight across correlated predictors (e.g., temp and atemp), reducing variance inflation. Lasso tends to **choose** one of a correlated pair and drop the other, which can improve interpretability but may be unstable if the correlated predictors contain nearly identical information.

Conclusion

This comprehensive analysis of the bike sharing dataset demonstrates the application of fundamental linear modeling techniques and diagnostic procedures.

Part I established a sensible baseline model with strong predictive power (adjusted $R^2 \approx 0.33$), identifying temperature, humidity, seasonal patterns, and system growth as primary drivers of bike rental demand.

Part II revealed violations of OLS assumptions, particularly heteroscedasticity, which were successfully addressed through log transformation of the response variable. While this transformation slightly reduced raw predictive performance, it improved diagnostic compliance and provides a more theoretically sound basis for statistical inference.

Part III identified severe multicollinearity between temperature and feeling temperature ($r = 0.992$, $VIF > 70$), demonstrating how collinearity inflates standard errors and destabilizes coefficient estimates without harming predictive accuracy. The recommended mitigation—removing `atemp`—successfully reduces VIF to acceptable levels while maintaining model performance.

Part IV evaluated out-of-sample predictive performance using 10-fold cross-validation and applied stepwise selection using AIC. The selected model was compared against the baseline using both in-sample fit (adjusted R^2) and cross-validated RMSE on the original rental scale.

Part V compared ridge regression, lasso regression, and the selected OLS model. Cross-validation results indicated that the selected OLS model achieved the best predictive accuracy (lowest CV RMSE on the original rental scale), while ridge and lasso provided principled ways to handle multicollinearity by shrinking coefficients (ridge) or selecting a sparse subset of predictors (lasso). Even when regularization did not outperform OLS in RMSE, it improved stability and interpretability in the presence of highly correlated predictors such as `temp` and `atemp`.

The log-transformed model with collinearity addressed serves as a strong, interpretable specification for predicting daily bike rentals.